HEATMAP NODE · PROJECT PLAN

HeatMap Node project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/11-plan.md @ b9c67bc		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First plan at the kit 1.9.1 upgrade; P4 to P6 shown as planned work only

HeatMap Node project plan

The repo is at stage P3 (TRL 3, constructable design, nothing built). The next gate is Amish's choice between option A and option B (HMN-DDR-004) and his review of this upgrade; the decision requested is in the last section. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem, users, requirements	Design lead	HMN-PRB-001, HMN-URS-001, HMN-PRS-001, HMN-RTM-001	Done at P3; to review with a co-design partner
STEM check and calculations	Design lead	HMN-STM-001, HMN-CAL-001	Done at P3; 15 rows plausible
Concept options	Design lead	HMN-OPT-001, HMN-DDR-004	Done; choice proposed, awaiting Amish
Detailed CAD (D3)	Design lead	`cad/src/model.py`, `cad/fidelity.json`, STEP files, drawing HMN-DWG-001	Done for options A and B
Build plan	Design lead	HMN-BLD-001	Done for option A (paper only)
Regulatory, risk, market	Design lead	HMN-REG-001, HMN-RSK-001, HMN-MKT-001, HMN-FEA-001	First pass done; items to assess
Partner and pole	Amish	Partner city, pole, attachment permission	Not started; nothing agreed
Renders	Coordinator	`media/options.png`, product renders	Rendered by the coordinator on Amish's Mac
Prototype build, field comparison	Partner	Built node, collocation with a reference WBGT meter	Planned (P4), on hold while the stage cap is P3

Resources needed

Table 2. Resources (for planned P4 work; estimates, nothing ordered).

Resource	Deliverable	Effort	Estimated cost (USD)
Parts for one head and FieldNode core	Built node	Collect and assemble over a few days	292.00 (`bom/bom.csv`)
Reference WBGT meter on loan	Collocated comparison	One partner site	To assess
Pole access and a lift or ladder with a second person	Fitting	Half a day	To assess
Sheet-metal forming and a printer	Saddle, shield and cowl	Workshop time	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 to P2	Problem, concept, requirements, media	Done
P3	Constructable design, calculations, STEM check, options, D3 CAD, build plan	Done; awaiting Amish's review and choice
P4	Prototype built and bench tested	Planned only
P5	Field comparison against a reference meter	Planned only
P6	Pilot network with a partner	Planned only

Next steps and decision requested

Decision requested from Amish: choose option A, option B or a hybrid (HMN-DDR-004, the scores 140 and 129 are within 10 %), and say whether to ask FieldNode for a cell charge cut-off (O4). Recommended next step: stop at TRL 3, review this upgrade, and if the design is wanted, decide whether the repo advances to TRL 4 once the portfolio cap allows.

References

- ISO 7243:2017, https://www.iso.org/standard/67188.html.
- Liljegren, J. C., R. A. Carhart, P. Lawday, S. Tschopp, and R. Sharp. "Modeling the Wet Bulb Globe Temperature Using Standard Meteorological Measurements." *Journal of Occupational and Environmental Hygiene* 5 (2008): 645 to 652. https://doi.org/10.1080/15459620802310770.
- US EPA. "Learn about Heat Islands." https://www.epa.gov/heatislands/learn-about-heat-islands.